

CRIZ (Admin IQ)

Design Phase Summary

Laabu AI Workforce · Phase 2 – Design

Item	Details
Project	Laabu AI Workforce
Agent	CRIZ (Admin IQ)
Document Type	Design Phase Summary
Lifecycle Phase	Phase 2 – Design
Status	Draft for Review
Architecture Direction	GCP first, cloud-neutral agent core

Executive Summary

This document explains how CRIZ will be built.

CRIZ is the operations backbone of the Laabu AI Workforce. It operates in two modes: an External Partner & Vendor Concierge and an Internal Operations & Workflow Assistant. As an external concierge, CRIZ handles vendor onboarding, compliance coordination, partner communications, contract collection, e-signature workflows, and documentation management. As an internal assistant, it supports contract review, meeting coordination, travel planning, procurement workflows, office operations, task tracking, executive assistance, approval routing, and operational reporting.

The first implementation runs on Google Cloud under the Laabu platform. The agent logic is kept separate from Google Cloud code. Models, storage, tools, secrets, identity, scheduling, and audit are reached through standard interfaces. This makes it possible to add AWS Bedrock or Azure AI Foundry later without rewriting the operational rules.

The most important safety rule is that the AI model never has final authority over sensitive actions. Approval checks, role validation, and compliance rules run in normal code, and any action that alters contracts, commitments, or spend thresholds requires explicit human approval.

1. Design Goals

The design must:

- support all CRIZ operational capabilities across both agent personas;
- demonstrate a complete vendor onboarding workflow end-to-end;
- maintain compliance controls and a full auditability trail;
- use approved Laabu context and shared event names;
- ensure cloud portability through standard interfaces;
- support role-based access control at every capability boundary;
- block sensitive actions without valid human approval;
- prevent duplicate operations during retries and replays;
- protect vendor, partner, and internal operational data;
- track cost and quality for every workflow run;
- run on GCP now while keeping future cloud options open.

2. Important Design Rules

1. No direct cloud SDK calls in business logic. Cloud-specific code lives in adapter folders only.
2. Human approval is required for sensitive actions. The approval must reference the exact action, record, and actor.
3. Normal code performs exact work. Approval checks, role validation, workflow state changes, compliance rules, scheduling calculations, budget thresholds, and task assignments use tested code, not model inference.
4. All external content is untreated as untrusted input. Documents, emails, vendor submissions, and tool responses are data, not instructions.

5. Every important step is recorded. The audit record includes the user, input, model level, tool, approval result, time, and cost.
6. Standard interfaces only. Modules communicate through typed contracts and may not directly mutate each other's internal state.
7. Least-privilege access controls. Each tool and role is granted only the permissions required for its defined purpose.

3. High-Level Architecture

The system is split into four layers.

Layer	Responsibilities
User & Event Layer	Receives chat requests, scheduled jobs, document uploads, inbound partner emails, and events from other Laabu agents such as AURA and FINA.
CRIZ Orchestration Layer	Validates identity and context, selects the active persona, chooses the operational capability, applies rules, and manages the full workflow lifecycle.
Tool Integration Layer	Connects to email, calendar, document, signature, procurement, scheduling, and workflow tools through approved MCP service contracts.
Control & Governance Layer	Applies identity checks, access rules, sensitive-data classification, human approval, audit recording, and monitoring across every request.

4. GCP Implementation Used in Version 1

Need	GCP Service Used in Version 1
Agent runtime	Vertex AI Agent Engine
Tool hosting	Cloud Run (MCP services)
Workflow state	Firestore
Documents and files	Cloud Storage
Secrets	Secret Manager
Large reports and queries	BigQuery
Identity and access	IAM and Agent Gateway controls
Sensitive-data checks	Sensitive Data Protection
Monitoring and audit	Cloud Logging, Trace, Monitoring, and Audit Logs

5. Cloud-Neutral Structure

The application core knows only about standard interfaces. Cloud-specific adapter code is isolated in the relevant adapter folder.

Interface	Simple Meaning
ModelProvider	Sends a request to an approved AI model and returns the response.
StateStore	Saves and reads workflow state records with version control.
ToolGateway	Calls an approved external tool using a standard request envelope.
IdentityContext	Tells the agent who the actor is, their role, and what they may do.
SecretStore	Reads a secret from the approved secret service at runtime only.
ObjectStore	Saves and reads documents, attachments, and generated files.
ApprovalService	Creates, checks, and records human approval decisions.
AuditSink	Records important workflow events to an append-only audit store.
Scheduler	Creates and manages reminder, follow-up, and recurring jobs.
Telemetry	Records traces, logs, latency, token counts, and cost per workflow.

The same interfaces are backed by different cloud adapters.

Need	GCP (Now)	AWS (Later)	Azure (Later)
AI models	Vertex AI	Bedrock	Azure AI Foundry
Runtime	Agent Engine	Approved container or agent runtime	Container Apps or approved agent runtime
State	Firestore	DynamoDB	Cosmos DB
Files	Cloud Storage	S3	Blob Storage
Secrets	Secret Manager	Secrets Manager	Key Vault
Identity	IAM	IAM	Entra ID and managed identity
Monitoring	Cloud Logging and Trace	CloudWatch and X-Ray	Azure Monitor and Application Insights
Tool hosting	Cloud Run	ECS, Lambda, or containers	Container Apps or Functions

Version 1 implements the GCP adapters. AWS and Azure remain future adapters with interface-level tests only.

6. CRIZ Orchestrator

The CRIZ orchestrator controls the full workflow lifecycle. It does not perform every task itself.

For each request, it:

8. validates the user, company, role, channel, and declared purpose;
9. loads the current session and workflow state;
10. retrieves only the approved operational context needed for the capability;
11. determines whether the request is external (partner or vendor) or internal (staff or system);
12. selects one operational capability module;
13. checks access, risk, and approval rules using exact code;
14. invokes exact code, an approved model step, or an approved tool;
15. saves the result, state version, and audit details;
16. returns the status, next required action, sources, and tracking ID.

A capability module is permitted to call only the tools explicitly listed for that module. Cross-module state access uses shared event contracts, not direct imports.

7. Operational Capability Modules

Module	Input	Main Work	Output
Vendor Onboarding	Vendor profile, documents, compliance category	Collect documents, validate fields, run compliance checks, route for approval, activate vendor record	Vendor status, compliance result, activation confirmation
Compliance Coordination	Compliance checklist, vendor record, policy version	Compare submitted documents against policy requirements, flag gaps, route exceptions	Compliance status, gap report, escalation task
Contract Lifecycle Support	Draft contract, counterparty details, approval chain	Review contract fields, collect e-signatures, track lifecycle stages, archive final version	Signed contract status, lifecycle stage, archive reference
Procurement Management	Purchase request, supplier, amount, budget code	Validate request against budget limits and policy, route for approval, issue purchase order	Approved PO, status, budget allocation update
Travel Coordination	Travel request, traveller details, dates, budget	Check policy compliance, find approved options, route for manager approval, track booking	Approved itinerary, booking confirmation, expense tracking ID
Meeting Management	Meeting request, attendees, agenda, platform	Check calendar availability, send invites, distribute agenda, record minutes, create follow-ups	Confirmed meeting, calendar

Module	Input	Main Work	Output
			entries, minutes document
Executive Assistance	Task request, priority, deadline, context	Triage request, prepare draft responses or summaries, route actions, schedule follow-ups	Completed task, draft output, next action
Task Tracking	Task details, assignee, due date, priority	Create task record, send assignments, monitor status, escalate overdue items	Task status, completion record, escalation log
Operational Reporting	Approved reporting scope, date range, audience	Query approved operational data, calculate summaries, explain results, cite sources	Formatted report with sources, run ID, freshness date

8. Common Request Flow

A failed step returns a clear status code and reason. The workflow must not stop silently.

Step	Action	Notes
1	Receive Request	Accept from chat, scheduled event, email trigger, or agent event.
2	Validate Input	Check format, required fields, and data classification.
3	Check Identity	Validate actor, company, role, and channel using exact code.
4	Load Context	Load session state and approved operational context only.
5	Select Capability	Route to the correct operational module using rules, not model inference.
6	Apply Rules	Run access, risk, compliance, and budget checks in code.
7	Call Approved Tool	Invoke tool through the ToolGateway with a standard envelope.
8	Request Approval if Needed	Pause workflow and create an approval record for sensitive actions.
9	Save State	Write versioned state with idempotency key and correlation ID.
10	Audit Event	Append event to the audit sink with actor, result, cost, and timestamp.
11	Return Result	Return status, tracking ID, next action, and sources.

9. First Vertical Slice

The first working slice proves the complete vendor onboarding story.

Stage	Action	Actor
1. New Vendor Request	Vendor submits onboarding request with required profile data.	External Vendor
2. Document Collection	CRIZ collects required compliance documents via document-mcp.	CRIZ + Vendor
3. Compliance Review	Compliance Coordinator reviews documents against active policy version.	Compliance Reviewer
4. Approval Required	CRIZ pauses workflow and creates an approval record.	CRIZ (automated)
5. Approved By	Authorised approver confirms vendor activation within their approval limit.	Approver
6. Contract Activation	CRIZ activates the vendor record and triggers contract workflow.	CRIZ (automated)
7. Audit Record Created	Full workflow trace is written to the audit sink.	Audit Service
8. Operational Status Returned	Vendor and internal teams receive activation confirmation.	CRIZ + LAABU

9.1 Success Criteria

- No duplicate vendor record is created during replay or retry.
- Document collection follows the active compliance policy version.
- Vendor activation is blocked without a valid approval record.
- Approval references the exact vendor record, documents submitted, and approval limit used.
- Retrying the same onboarding event does not create a second vendor record.
- Every workflow step is traceable through a single correlation ID.

10. Detailed Workflow Rules

10.1 Vendor Onboarding

17. Read and validate the vendor profile fields using exact code.
18. Stop if a vendor record with the same registration number already exists.
19. Collect required documents through document-mcp using the active policy checklist.
20. Store extracted fields and confidence scores for each document.
21. Flag missing documents or low-confidence extractions for compliance review.
22. Route the compliance review task to the Compliance Reviewer role.
23. Pause workflow and create an approval record before activating the vendor.
24. Record the approval result, approver identity, and timestamp before proceeding.

10.2 Contract Management

25. Read draft contract fields through document-mcp.
26. Validate counterparty details, effective dates, and required clauses.
27. Route the contract for e-signature through signature-mcp in the defined order.
28. Block contract activation until all required signatories have completed signing.
29. Archive the final signed document to object storage with a version reference.
30. Update the contract lifecycle stage in state and emit a contract_status event.

10.3 Travel Requests

31. Read traveller details, travel dates, destination, and purpose.
32. Check the request against active travel policy limits using exact code.
33. Reject or flag requests that exceed policy limits without escalation.
34. Route for manager approval with the full request detail and policy check result.
35. Confirm booking only after approval is recorded.
36. Track booking confirmation and emit the travel_requested event.

10.4 Procurement Requests

37. Read purchase request fields including supplier, amount, currency, and budget code.
38. Validate against active budget limits and procurement policy using exact code.
39. Require dual approval for requests above the defined spend threshold.
40. Issue a purchase order only after all required approvals are recorded.
41. Emit the procurement_requested event and update the budget allocation record.

10.5 Meeting Coordination

42. Check all required attendee calendars through calendar-mcp for conflicts.
43. Send meeting invitations with agenda through email-mcp.
44. Record the meeting in the meetings state collection.
45. Distribute minutes and create follow-up tasks after the meeting.

10.6 Task Management

46. Validate task fields including assignee, priority, deadline, and description.
47. Create the task record and send assignment notification through email-mcp.
48. Monitor task status and escalate overdue tasks to the responsible manager.
49. Record task completion with evidence and close the tracking record.

11. Tools and Service Contracts

MCP is the standard connection between CRIZ and each tool. It keeps tool implementation separate from agent logic.

Tool or Service	Purpose	Controls	Permissions
laabu-agent-gateway	Entry point for all requests: identity, policy, and routing.	Every request is authenticated and tenant-scoped.	Read identity context; enforce routing policy.
session-mcp	Session and workflow state management.	Tenant-scoped access; version-controlled writes.	Read and write own tenant state only.
email-mcp	Vendor and partner communications; internal notifications.	Approved templates; DLP scan before send; send audit.	Send to verified recipient list; no bulk send.
calendar-mcp	Meeting scheduling, availability checks, and calendar management.	Attendee consent required; no write without invite.	Read availability; create calendar events.
document-mcp	Read vendor documents, contracts, and generated reports.	File type, size, and confidence threshold checks.	Read from approved storage paths only.
signature-mcp	E-signature workflow management for contracts and approvals.	Signer identity verified; audit trail required.	Create and track signature requests only.
procurement-mcp	Issue and track purchase orders and procurement workflows.	Budget validation before issue; approval required.	Create PO within approved limits only.
workflow-mcp	Approval routing and workflow state management.	Role-based routing; escalation rules enforced.	Create and update workflow tasks.
scheduler	Reminder, follow-up, and recurring job management.	Job ownership validated; duplicate prevention enforced.	Create and cancel own jobs only.
audit-service	Append-only workflow evidence recording.	No user or agent can edit or delete audit records.	Append only; no read-back for business logic.

Every tool request uses a standard envelope containing: `tenant_id`, `actor_id`, `request_id`, `correlation_id`, `idempotency_key`, `purpose`, `data_classification`, `timeout`, and `validated_payload`. A tool response is treated as data. It cannot modify the agent's rules, routing logic, or access decisions.

12. Model Strategy

CRIZ uses model capability levels, not hard-coded model names. Changing the approved model requires a configuration change followed by evaluation and cost checks.

Model Level	Used For	Not Used For
Workhorse	Request routing, intent classification, scheduling summaries, draft communications, short field extractions.	Final approval decisions, compliance rule execution, or contract authority.
Reasoning	Complex workflow decisions, compliance analysis, multi-document contract review, operational reporting with cited sources.	Direct tool invocation, budget decisions, or role validation.

13. Exact Logic Boundary

The AI model may explain or summarise, but the following must always use normal code or deterministic logic:

- approval checks and approval limit validation;
- role and permission validation;
- workflow state transitions;
- compliance rule application;
- scheduling and deadline calculations;
- budget threshold checks;
- task assignment and escalation rules;
- idempotency and duplicate prevention decisions;
- vendor or contract record activation.

This boundary makes the system auditable, testable, and legally defensible.

14. State and Event Contracts

14.1 Shared Events

Event Name	Meaning
vendor_submitted	A vendor has submitted an onboarding request with required profile data.
compliance_review_required	A compliance review task has been created and is awaiting a reviewer.
approval_required	A sensitive action is paused and awaiting an authorised approver.
approved_by	An authorised person has approved the exact action with recorded limit.
contract_activated	A contract has been signed by all parties and archived.
travel_requested	A travel request has passed policy checks and is tracking booking status.

Event Name	Meaning
procurement_requested	A purchase request has been approved and a purchase order has been issued.

14.2 Key State Fields

Field	Purpose
tenant_id	Keeps company data strictly separate across all collections.
actor_id	Identifies the person or service making the request.
workflow_id	Identifies the specific operational workflow instance.
correlation_id	Links events, tool calls, and log entries across the full workflow.
persona	External (vendor or partner) or internal (staff or system) mode.
intent	The requested operational capability selected by routing logic.
risk_level	Low, medium, or high, determined by exact rules in code.
approval_id	Reference to the human approval record for this action.
source_refs	Documents and records used to support the workflow action.
idempotency_key	Prevents duplicate records during retries or replayed events.
next_action	The next required step in the workflow, stored for recovery.

Session state does not store secrets, full contract text, or full vendor documents. These remain in approved object storage with a reference key only.

15. Storage Design

Proposed Firestore collections:

Collection	Purpose
vendors	Vendor profile, status, compliance result, and activation record.
contracts	Contract lifecycle stage, signatories, signed references, and archive pointer.
travel_requests	Travel request fields, policy check result, approval, and booking confirmation.
procurement_requests	Purchase request, approval chain, PO reference, and budget allocation.
meetings	Meeting details, attendees, agenda, minutes, and follow-up tasks.

Collection	Purpose
tasks	Task assignments, status, due dates, completion evidence, and escalation history.
approvals	Approval records with approver identity, limit, timestamp, and approval hash.
audit_events	Append-only workflow evidence with full actor, action, cost, and result detail.
reports	Saved operational report runs, sources used, and freshness timestamps.

15.1 Storage Principles

- Every record includes: tenant_id, workflow_id, status, source system, created time, updated time, version, and idempotency key.
- Documents are stored as object storage references. Raw file content is not stored in Firestore.
- Secrets are stored only in the approved secret service. No secret values in state or audit records.
- Audit records are append-only. No agent or user role may modify or delete an existing audit event.

16. Access Control and Human Approval

Access uses both role and business context. A role alone is not sufficient. The system also checks company, department, vendor, contract, amount, and environment.

User Type	Main Access
External Vendor	Only authorised onboarding forms, document upload, and own vendor status.
Operations Coordinator	Vendor records, meeting management, task tracking, and procurement requests.
Compliance Reviewer	Vendor compliance documents, compliance checklists, and gap reports.
Executive Assistant	Executive task queue, meeting management, travel requests, and draft communications.
Manager	Team task assignments, travel and procurement approvals within defined limits.
Approver	Approve sensitive actions within their registered approval limit only.
Auditor	Read-only access to audit events, approval records, and workflow history.

16.1 Approval Requirements

Human approval records must store:

- the identity of the approver and their registered approval limit;
- the timestamp of the approval decision;
- the exact action, record reference, and data used for the approval;
- a hash of the source data at the time of approval;
- an expiry time after which the approval is no longer valid;
- the approval or rejection reason.

If any protected field changes after approval is recorded, the approval becomes invalid and a new approval must be requested.

17. Security and Trust Boundaries

CRIZ follows these trust rules:

- External documents, emails, vendor submissions, and tool responses are treated as untrusted data at all times.
- Hidden instructions embedded in documents or emails are ignored. Tool calls are constructed from validated fields only.
- Tool calls are built from validated, structured fields. Raw model output is never passed directly to a tool.
- Secrets are read only at runtime from the approved secret service. No secret values are stored in prompts, logs, or session state.
- Sensitive field values are removed or masked from prompts and log outputs where possible.
- All external responses pass a sensitive-data scan before inclusion in any output.
- Every query and tool call includes the tenant scope. Cross-tenant access is not possible through the agent layer.
- Tool permissions follow least privilege. Each MCP service has only the permissions needed for its defined purpose.
- Audit records use append-only storage. No normal role may edit or delete an existing audit event.
- All high-risk and sensitive actions require human approval before execution.

18. Reliability and Failure Handling

18.1 Duplicate Prevention

Repeating the same request must not create a second vendor record, contract, PO, or approval. Each write operation uses an idempotency key. The existing record is checked before any create operation.

18.2 Retry Policy

Failure	Response
Temporary network error before any action	Retry with a short exponential backoff.
Tool rate limit	Wait and retry within a defined small limit.
Validation error	Do not retry. Return a clear error with the specific field that failed.
Approval timeout	Escalate to the next approver in the defined chain.
Low document confidence	Route to human review rather than making an inference.
Missing source or access	Stop and request the required input or permission explicitly.

18.3 Human Fallback

When confidence is low, data conflicts, or a high-risk rule produces an unclear result, CRIZ creates a review task instead of guessing. The review task includes the full workflow context, the specific conflict, and the next action required to unblock the workflow.

19. Observability and Cost Tracking

Every workflow records:

- workflow ID and correlation ID;
- tenant ID and actor ID;
- selected operational capability module;
- model level and resolved model ID;
- input and output token counts;
- tool name and tool call status;
- approval status and approval ID;
- end-to-end latency;
- document count and attachment references;
- estimated cost per workflow run;
- error code and error message when applicable;
- final outcome and next action.

Main dashboards should show: workflow success rate, error rate, approval rate, duplicate prevention events, vendor activation rate, contract cycle time, document confidence distribution, cost per workflow, and end-to-end response latency.

20. Testing and Evaluation

Test Level	What It Checks
Unit Tests	Approval rules, role validation, routing logic, state transitions, scheduling calculations, and duplicate detection.
Contract Tests	Event names, tool request and response schemas, and storage record formats.
Integration Tests	GCP adapters, MCP tool connections, and Firestore and Cloud Storage operations.
Agent Evaluations	Intent routing accuracy, grounded responses, source citation, and end-to-end workflow quality.
Security Tests	Access bypass attempts, prompt injection resistance, data leakage checks, and approval bypass attempts.
Portability Tests	Application core contains no cloud SDK imports; all adapters implement the same standard interfaces.

20.1 Initial Pass Targets

- Workflow intent routing: at least 95%.
- Required document field extraction: at least 95%.
- Exact rule calculations (approval limits, budget thresholds): 100%.
- Operational reports with cited sources: 100%.
- Unauthorised access requests refused: 100%.
- Sensitive actions blocked without valid approval: 100%.
- Duplicate vendor or contract records during replay: 0.
- Required sensitive-data removal from outputs: 100%.

21. Folder Structure

The project structure separates domain rules, application logic, cloud adapters, and capability modules. Cloud-specific imports are allowed only inside the relevant adapter folders.

```
agents/
└─ AdminIQ/
   └─ AdminIQ_orch/
      └─ orchestrator_agent.py
   └─ AdminIQ_agent/
      └─ docs/
         ├── Planning.md
         ├── Design.md
         ├── Budget.md
         ├── Cost_report.md
         ├── sample_prompts.md
         └─ Demo.md
      └─ implementation/
         ├── domain/
         ├── application/
         ├── interfaces/
         ├── adapters/
         │   ├── gcp/
         │   ├── aws/
         │   └─ azure/
         └─ capabilities/
            ├── vendor_onboarding/
            ├── compliance/
            ├── contracts/
            ├── procurement/
            ├── travel/
            ├── meetings/
            ├── executive_assistance/
            ├── task_tracking/
            └─ reporting/
      └─ config/
      └─ memory/
      └─ scheduler/
      └─ evaluation/
      └─ tests/
      └─ deploy/
      └─ pyproject.toml
```

22. Development Approach

Build CRIZ in small working steps. Each step must produce a testable, working capability before the next begins.

Each capability module should define:

- typed input and output schemas;
- the list of allowed tools for that module;
- the model level required for each step;
- exact rules that must run in code;
- side effects and state changes;
- error handling and fallback behaviour;
- test cases and evaluation prompts;
- cost and quality metrics.

Modules communicate through typed event contracts. They must not directly import or mutate each other's internal state.

The normal branch path is: feature branch → develop → release → security checks → main.

The recommended build order is:

50. Approve planning, design, and budget documents.
51. Create the standard project skeleton with typed schemas and interfaces.
52. Build fake tool implementations for local testing.
53. Build the vendor onboarding and compliance flow end-to-end.
54. Add the contract lifecycle and e-signature workflow.
55. Add procurement, travel, and meeting coordination modules.
56. Add task tracking, executive assistance, and operational reporting.
57. Connect approved GCP adapters and replace fake tools.
58. Add tests, evaluations, security checks, and cost tracking.
59. Run the pilot and improve the design from real operational results.

23. Deployment Gates

Before production deployment, the following checks must pass:

- Planning, Design, and Budget documents are approved.
- Security scan has no blocking findings.
- Actual cost estimates are compared with the approved budget.
- Agent evaluations meet all required pass targets.
- Budget alerts, spend caps, and cost labels are active and confirmed.
- All required MCP service contracts have passing contract tests.
- Human approval flow is tested end-to-end with audit verification.
- Portability tests confirm no cloud SDK imports in the application core.

24. Definition of Done

CRIZ Version 1 is complete when:

- it runs under the approved Laabu orchestration platform;
- all nine operational capability modules have a working path or approved staged plan;
- the first end-to-end vendor onboarding story works and is demonstrable;
- operational reports include cited sources and a freshness timestamp;

- access control, sensitive-data checks, human approval, and audit are active;
- approval and compliance rules pass all defined test cases;
- duplicate vendor, contract, and procurement records are prevented during replay;
- agent evaluations pass all required targets;
- cost and latency are visible in dashboards for every workflow run;
- documentation, demonstration steps, and a runbook are complete;
- the actual cost report is submitted against the approved budget;
- the GCP version runs end-to-end and the future provider interfaces pass contract tests.

25. Open Decisions

The following items must be confirmed before live integration:

- final approved model IDs for the workhorse and reasoning levels;
- exact shared framework folder and class names aligned with other Laabu agents;
- first e-signature provider and signature-mcp contract version;
- first procurement connector and purchase order format;
- approved compliance policy document format and versioning scheme;
- vendor_submitted and contract_activated shared event payload schemas;
- approval limit values and expiry rules per role and action type;
- Firestore tenant and environment partition layout;
- document retention periods for contracts, vendor records, and audit events;
- regional and data-residency settings for the GCP deployment.